Programming Fundamentals – COC1071
Lab Plan – 11

Learning Objective:

- *Arrays*
- *Passing Arrays to Functions*

1) Write a program to find the largest element in an array of 10 integers.

2) Write a program to count the frequency of each element in an array.

3) Write a program to sort an array of integers in ascending order without using built-in functions.

4) Write a program to find the position of a specific element in an array (linear search).

5) Write a function that prints all elements of an array along with their indices.

6) Create a program to find the sum of all positive numbers and all negative numbers separately in an array.

7) Create a function that computes the sum of all elements in an integer array and returns the result. Call this function from the main program.

8) Implement a function that counts the number of even numbers in an array and returns the count.

9) Write a function that takes an array and its size as arguments and reverses the array elements in place.

10) Write a function that merges two arrays into one sorted array and returns the resulting array to the caller.

11) Implement a function to count how many times a given value appears in an array.

12) Write a function to find and return the sum of all prime numbers in an array.

13) Create a function to split an array into two arrays: one containing all positive numbers and the other containing all negative numbers.

14) Write a function that finds the median of an array. Assume the array is unsorted.

15) Create a function that multiplies two arrays element-wise and returns the resulting array.